

Jongmin Choi

M.S. Student in Electrical Engineering · Multimodal AI (MMAI) Lab, KAIST

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RESEARCH INTERESTS

Biological intelligence is the clearest existence proof we have of a general and efficient learner. I believe that networks shaped by its principles rather than by scale alone can become more capable and efficient. My master's research followed this belief in two directions. First, I studied the gap between audio and text and how multimodal models can narrow it while preserving the structure unique to each modality. Second, I asked what the basic unit of computation should be. A biological neuron is not simply a perceptron. It spikes, resets, and is shaped by its membrane channels. I developed spiking neuron algorithms that place these mechanisms within the neuron itself. I am now studying connectors between audio encoders and language-model decoders to reduce information bottlenecks. In parallel, I investigate the mathematical foundations of optimization algorithms for training deep networks.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea Mar 2025 – Present
M.S. in Electrical Engineering

Advisor: **Prof. Joon Son Chung** · Multimodal AI (MMAI) Lab · GPA 4.06 / 4.30

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea Mar 2018 – Feb 2025
B.S. in Electrical Engineering and Mathematical Sciences (Double Major)

GPA 3.65 / 4.30

PUBLICATIONS

[C] Conference · [P] Preprint · * Equal contribution

[P] **OmniKVQuant: KV Cache Quantization for Omni-LLMs**

Suho Yoo*, Hyunjong Ok*, **Jongmin Choi**, Jihoo Jung, Joon Son Chung

Submitted to *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)* · 2027 UNDER REVIEW

[P] **FiTS: Interpretable Spiking Neurons via Frequency Selectivity and Temporal Shaping**

Jongmin Choi, Joon Son Chung

Submitted to *Neural Information Processing Systems (NeurIPS)* · 2026 UNDER REVIEW

[C] **LAMB: LLM-based Audio Captioning with Modality Gap Bridging via Cauchy-Schwarz Divergence**

Hyeongkeun Lee*, **Jongmin Choi***, KiHyun Nam, Joon Son Chung

IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) · 2026 ORAL

[C] **Diffusion-Link: Diffusion Probabilistic Model for Bridging the Audio-Text Modality Gap**

KiHyun Nam*, **Jongmin Choi***, Hyeongkeun Lee, Jungwoo Heo, Joon Son Chung

IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) · 2026 POSTER

[C] **Fork-Merge Decoding: Enhancing Multimodal Understanding in Audio-Visual Large Language Models**

Chaeyoung Jung*, Youngjoon Jang*, **Jongmin Choi**, Joon Son Chung

Findings of Empirical Methods in Natural Language Processing (EMNLP) · 2026

TEACHING EXPERIENCE

EE.70038 — Speech Recognition Systems — KAIST Spring 2026

Teaching Assistant · Instructor: **Prof. Joon Son Chung**

EE.40034 — Deep Learning for Visual Understanding — KAIST Fall 2025

Teaching Assistant · Instructor: **Prof. Joon Son Chung**

SKILLS

Research & ML: Python, PyTorch · **iOS Development:** Swift, SwiftUI, SwiftData · **Languages:** Korean (native), English (professional working proficiency)

Last updated: September 2026